CIVICSENSE
PUBLIC GRIEVANCE & SENTIMENT ANALYSIS

Final Project Review | Department of Computer Science & Engineering | K L (Deemed to be) University

TEAM MEMBERS

2420030098– B.Chetna

2420030262 – J.Nandini

2420030266– S.Saranya

Guide: Dr.N.Sunanda | Academic Year: 2026–2027

Introduction & Problem Statement

Background

Public grievance portals collect free-text complaints about roads, water, electricity, healthcare and other civic services. NLP can convert these text records into structured insights.

Problem Statement

Manual sorting of unstructured complaints is slow and inconsistent. It is difficult to identify service categories, citizen sentiment and recurring patterns at scale.

Motivation / Need

Civic analysts and service teams need faster summaries to understand recurring issues and public dissatisfaction. CivicSense supports analysis; it does not replace official grievance resolution.

CITIZEN GRIEVANCE

Free-text complaint

↓

Category + sentiment

↓

Trends & insights

Decision-support view for civic analysts

Objectives & Scope

- **Collect, clean and preprocess** government grievance data by handling missing values, duplicate records and unnecessary information.
- **Apply NLP and Machine Learning** to process citizen complaints and classify them into categories such as Roads, Water Supply, Electricity, Healthcare, Education and Transportation.
- **Perform Sentiment and Trend Analysis** to identify Positive, Negative and Neutral feedback and understand complaint patterns over time.
- **Develop an interactive Streamlit dashboard** to visualize complaint categories, sentiment distribution, trends and key insights.

Literature Survey

Authors	Title of the Paper	Methodology	Inferences	Limitation/Future work
Pang, Lee & Vaithyanath an	Thumbs up? Sentiment Classification using Machine Learning Techniques	Naive Bayes, Maximum Entropy, SVM	Machine learning can effectively classify text into sentiment categories.	Mainly focuses on movie reviews; government grievance-specific analysis can be explored.
Medhat, Hassan & Korashy	Sentiment Analysis Algorithms and Applications: A Survey	• Lexicon-based and Machine Learning approaches	• Provides an overview of different sentiment analysis techniques and applications.	• Sentiment analysis can be affected by domain and language complexity.
Devlin et al	BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding	• Transformer-based NLP	• Contextual language models provide strong performance in text understanding.	• Requires higher computational resources compared with basic ML approaches.

Proposed System & Methodology

STEP 1

Define Problem

Define the grievance-analysis need, users and project scope.

STEP 2

Inspect Data

Use the supplied CSV demo dataset: 80 rows, 9 fields.

STEP 3

Design Models

Plan preprocessing, TF-IDF, Logistic Regression and analytics.

STEP 4

Build Modules

Develop React UI, FastAPI services and Streamlit dashboard.

STEP 5

Test & Evaluate

Use test cases and classification metrics; record real scores after execution.

Key Novelty / Contribution

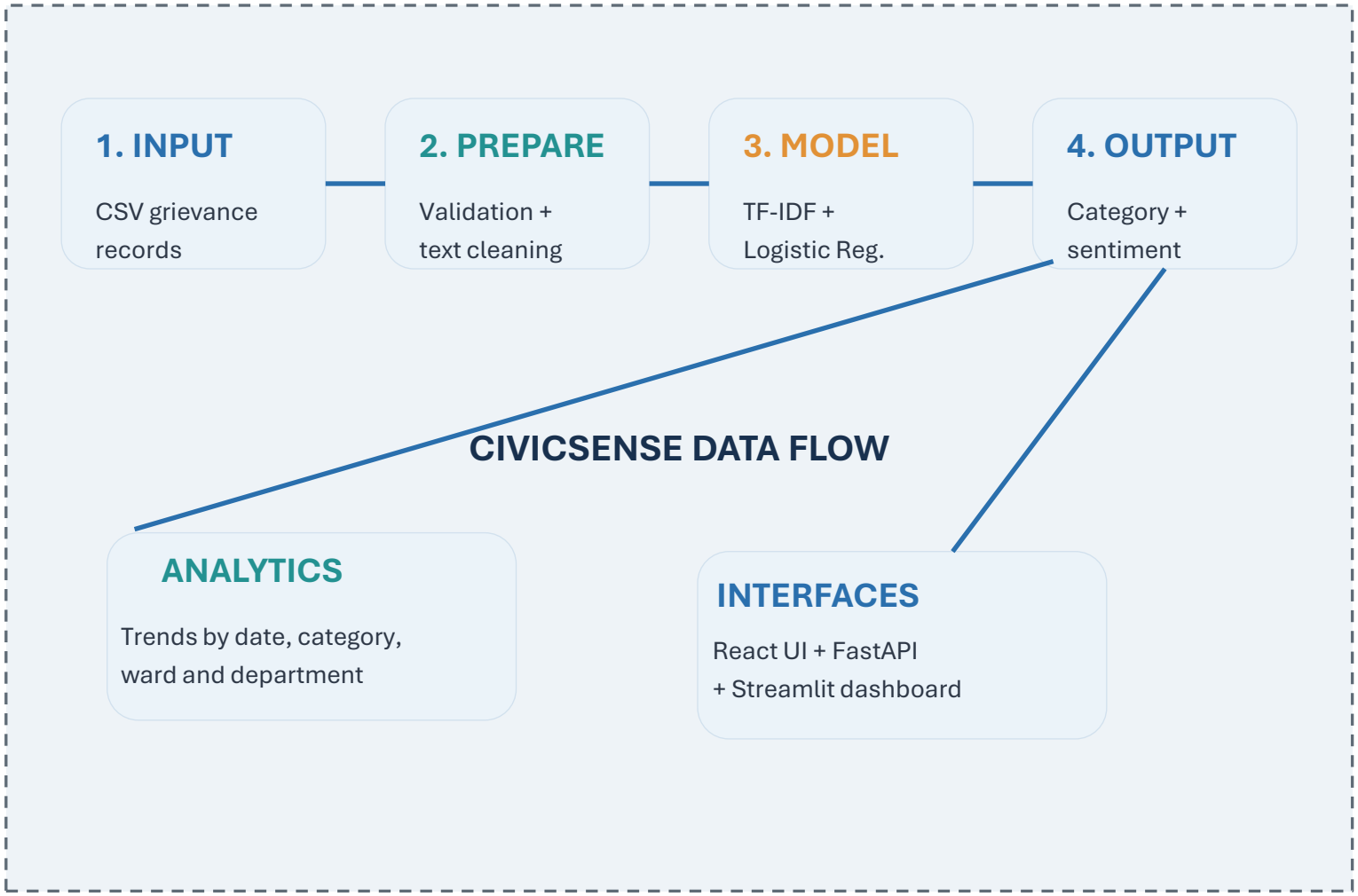
Contribution: CivicSense unifies complaint categorization, sentiment prediction and interactive civic analytics using a lightweight, interpretable NLP pipeline.

Proposed System Architecture

1. Data Acquisition
Collect government grievance records and labeled sentiment data.
Sources may include CPGRAMS or publicly available state grievance portals.
2. Data Cleaning
Handle missing values, duplicates, irrelevant fields, and inconsistent formats.
Prepare a clean dataset for analysis.
3. NLP & Feature Extraction
Clean and normalize complaint text.
Apply **TF-IDF** to convert text into numerical features.
4. Machine Learning Analysis
Complaint Classification: Categorize complaints into roads, water supply, electricity, healthcare, education, transportation, and other services.
Sentiment Analysis: Classify feedback as positive, negative, or neutral.

Technology Stack: Python, Pandas, NumPy, Scikit-learn, NLP

System Architecture



Components

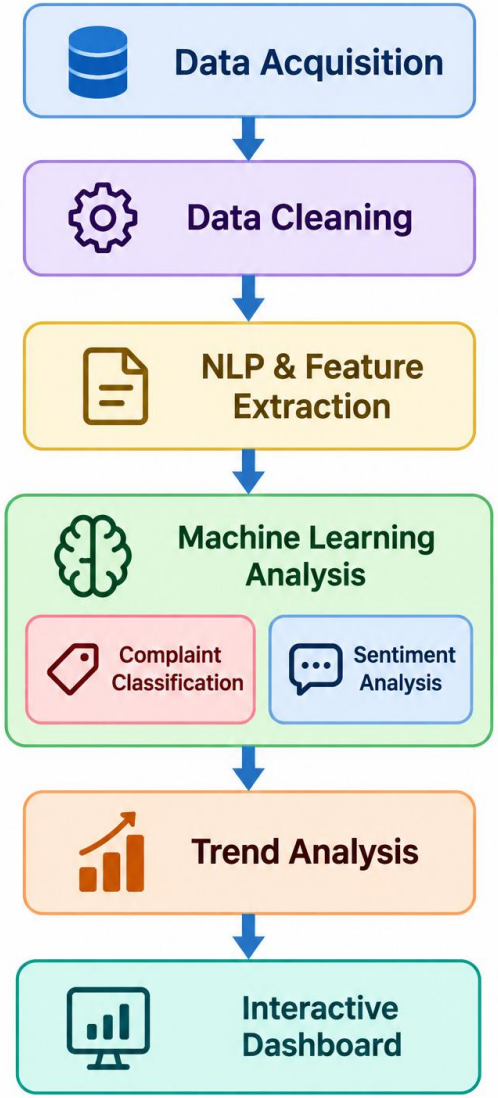
1. Data loader
   Reads CSV and validates fields
2. NLP pipeline
   Cleans and vectorizes text
3. ML models
   Predict category and sentiment
4. Analytics layer
   Builds trends and insights

5. Trend Analysis
Calculate daily, weekly, and monthly complaint counts.
Identify frequently reported service categories.
Analyze sentiment distribution and changes over time.

6. Interactive Dashboard
KPI Cards: Total grievances and sentiment counts.
Bar Chart: Complaints by service category.
Donut Chart: Positive, negative, and neutral sentiment.
Line Chart: Complaint trends over time.
Filters & Complaint Explorer: Search records by date, category, and sentiment.

Technology Stack: Streamlit, Plotly, Matplotlib

Expected Outcome: Transform grievance records into understandable insights that help identify recurring public-service issues.



Hardware & Software Requirements

Hardware

- Windows / Linux / macOS
- 8 GB RAM recommended
- Dual-core or better CPU
- Local storage + internet for setup

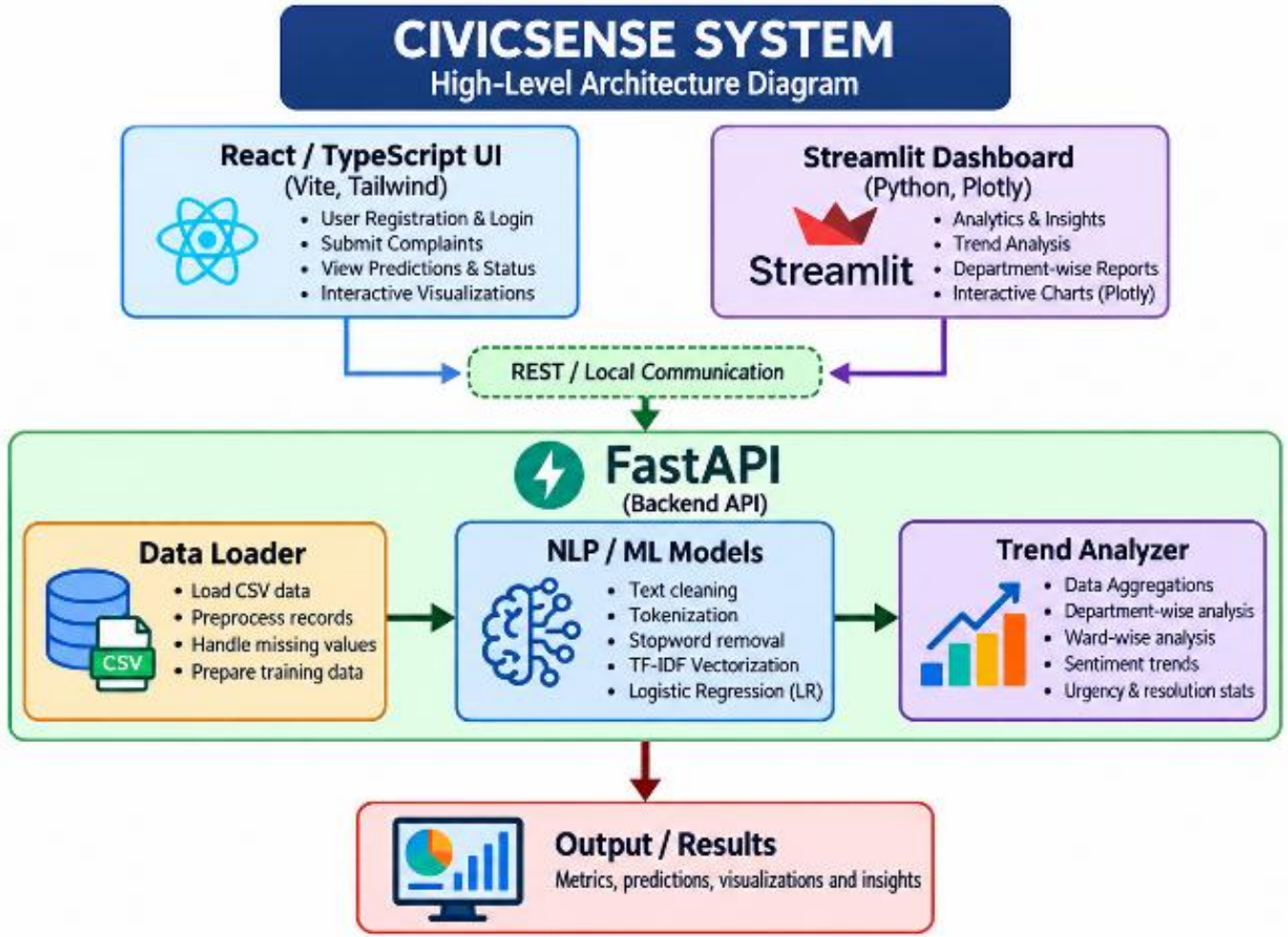
Software & Tools

- Python 3.x
- Node.js + npm
- VS Code
- Git / GitHub (if used)

Libraries / Frameworks

- Pandas, NumPy
- Scikit-learn
- FastAPI, Streamlit
- React, TypeScript, Vite
- Plotly, Matplotlib, Recharts

Deployment / Platform: Local development – Vite frontend, Uvicorn FastAPI backend, Streamlit dashboard. No cloud deployment evidenced.


CIVICSENSE SYSTEM
High-Level Architecture Diagram
React / TypeScript UI
(Vite, Tailwind)
User Registration & Login
Submit Complaints
View Predictions & Status
Interactive Visualizations
Streamlit Dashboard
(Python, Plotly)
Streamlit
Analytics & Insights
Trend Analysis
Department-wise Reports
Interactive Charts (Plotly)
REST / Local Communication
FastAPI
(Backend API)
Data Loader
CSV
Load CSV data
Preprocess records
Handle missing values
Prepare training data
NLP / ML Models
Text cleaning
Tokenization
Stopword removal
TF-IDF Vectorization
Logistic Regression (LR)
Trend Analyzer
Data Aggregations
Department-wise analysis
Ward-wise analysis
Sentiment trends
Urgency & resolution stats
Output / Results
Metrics, predictions, visualizations and insights

Dataset / Data Understanding

80

Records (rows)

9

Features (columns)

0

Missing values

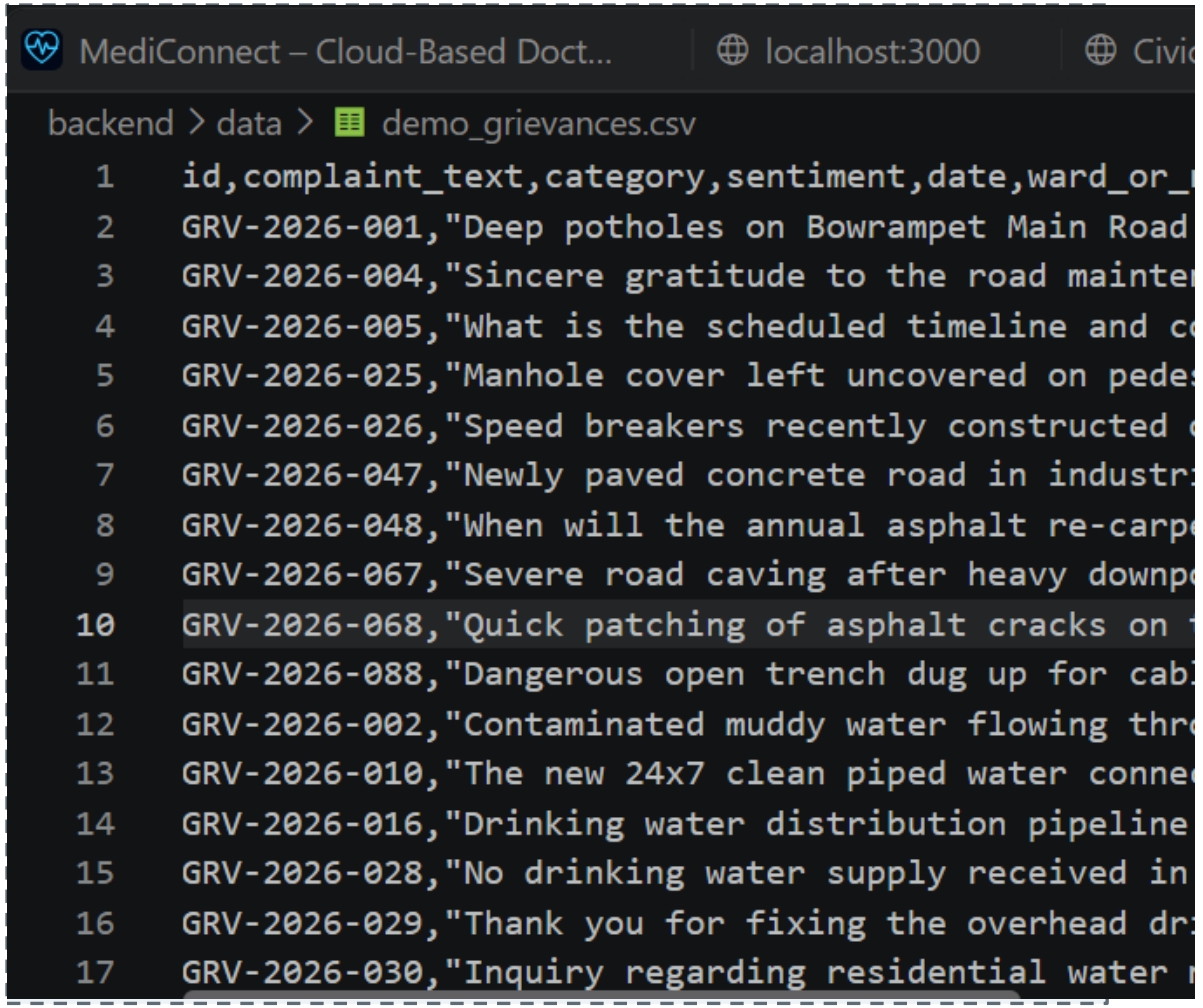
8

Target classes

- Source: Repository-provided demo CSV (backend/data/demo_grievances.csv)

Fields: complaint_text, category, sentiment, date, ward/region, department, status, urgency and ID.

Quality: 0 missing cells; 0 exact duplicate rows in the supplied CSV. 8 service categories.



Data Preprocessing & Exploratory Analysis

CATEGORY DISTRIBUTION

8 service categories

10 demo records per category

Roads • Water • Electricity

Healthcare • Education

Transport • Drainage • Other

TEXT PROCESSING FLOW

Raw complaint text

↓

Lowercase + remove URLs/email

↓

Remove punctuation / digits

↓

Tokenize + stopword removal

↓

TF-IDF unigram + bigram vectors

Cleaning

No missing cells or exact duplicate rows in supplied CSV. Cleaning functions also handle blank text and repeated rows.

Transformation

Normalize text and convert it into sparse TF-IDF numerical features. No numeric scaling required for text vectors.

Feature Selection

TF-IDF with 1–2 grams; max 1,500 features for category and 1,200 for sentiment in backend models.

Module-wise Description

1 Data Management

Purpose: Load, validate and clean grievance CSV.
Input → Output: CSV → quality-checked records.
Developed by: [Member name]

2 NLP & Classification

Purpose: Clean text and predict service category.
Input → Output: Complaint → 8-class label.
Developed by: [Member name]

3 Sentiment Analysis

Purpose: Detect positive, negative or neutral tone.
Input → Output: Complaint → sentiment label.
Developed by: [Member name]

4 Analytics & API

Purpose: Trends, insights, dashboard and API.
Input → Output: Records → charts / JSON.
Developed by: [Member name]

Algorithms and techniques

1. TF-IDF (Term Frequency–Inverse Document Frequency)

TF-IDF converts complaint descriptions into numerical features that machine-learning models can understand. It assigns importance to words based on their frequency in a complaint and how common or rare they are across the dataset.
Purpose: Text feature extraction.
Example: Words such as “pothole,” “drainage,” and “electricity” may help distinguish complaint categories.

2. Logistic Regression

Logistic Regression is a supervised machine-learning algorithm used to predict a class based on input features.
It is used for classification.
Purpose: Predict complaint categories and sentiment labels.
Complaint classification: Predicts one of eight categories, including Roads, Water Supply, Electricity, Healthcare, Education, Transportation, Drainage / Sewage, and Other Public Services.

Sentiment analysis: A separately trained model predicts Positive, Negative, or Neutral sentiment.
Advantage: Efficient, interpretable, and suitable as a baseline for TF-IDF text classification.

3. Train-Test Split

Train-test split divides labeled data into two parts:
One for **training the model** and another for **evaluating its performance** on unseen examples.

- Training set: Used to fit the TF-IDF vectorizer and train Logistic Regression.
- Testing set: Used to measure how well the trained model generalizes to unseen complaints.
- Purpose: Helps evaluate the model fairly and identify overfitting.

4. Data Aggregation for Trend Analysis

Data aggregation groups complaint records according to fields such as date, category, and sentiment, then calculates summary statistics.

- Calculates daily, weekly, and monthly complaint counts.
- Identifies frequently reported service categories.
- Examines sentiment distribution across categories.
- Tracks changes in complaint frequency over time.

Example: Counting complaints each month to examine whether their frequency changes.

Implementation / Model Building

Feature Selection

Input feature: complaint_text. Target labels: category and sentiment. Other columns support dashboard analytics.

Preprocessing Pipeline

Lowercase → remove URLs/email → punctuation cleanup → tokenize → stopword removal → TF-IDF.

Data Splitting

Python backend default: 75% training / 25% testing. Stratification is attempted when class counts permit.

Configuration

Logistic Regression: max_iter=1000; balanced class weights. TF-IDF: unigram + bigram; capped vocabulary.

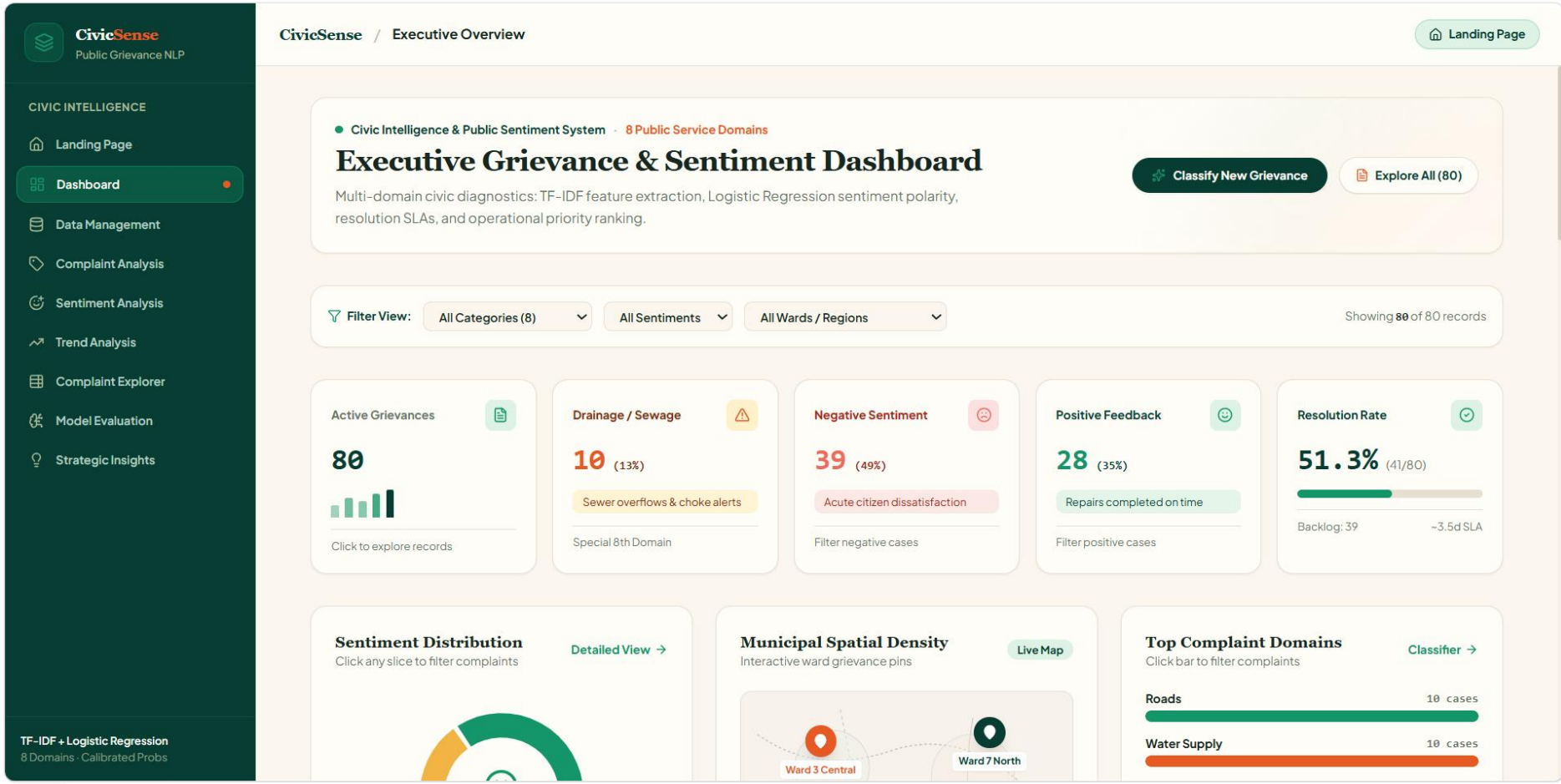
Code snippet / pseudocode

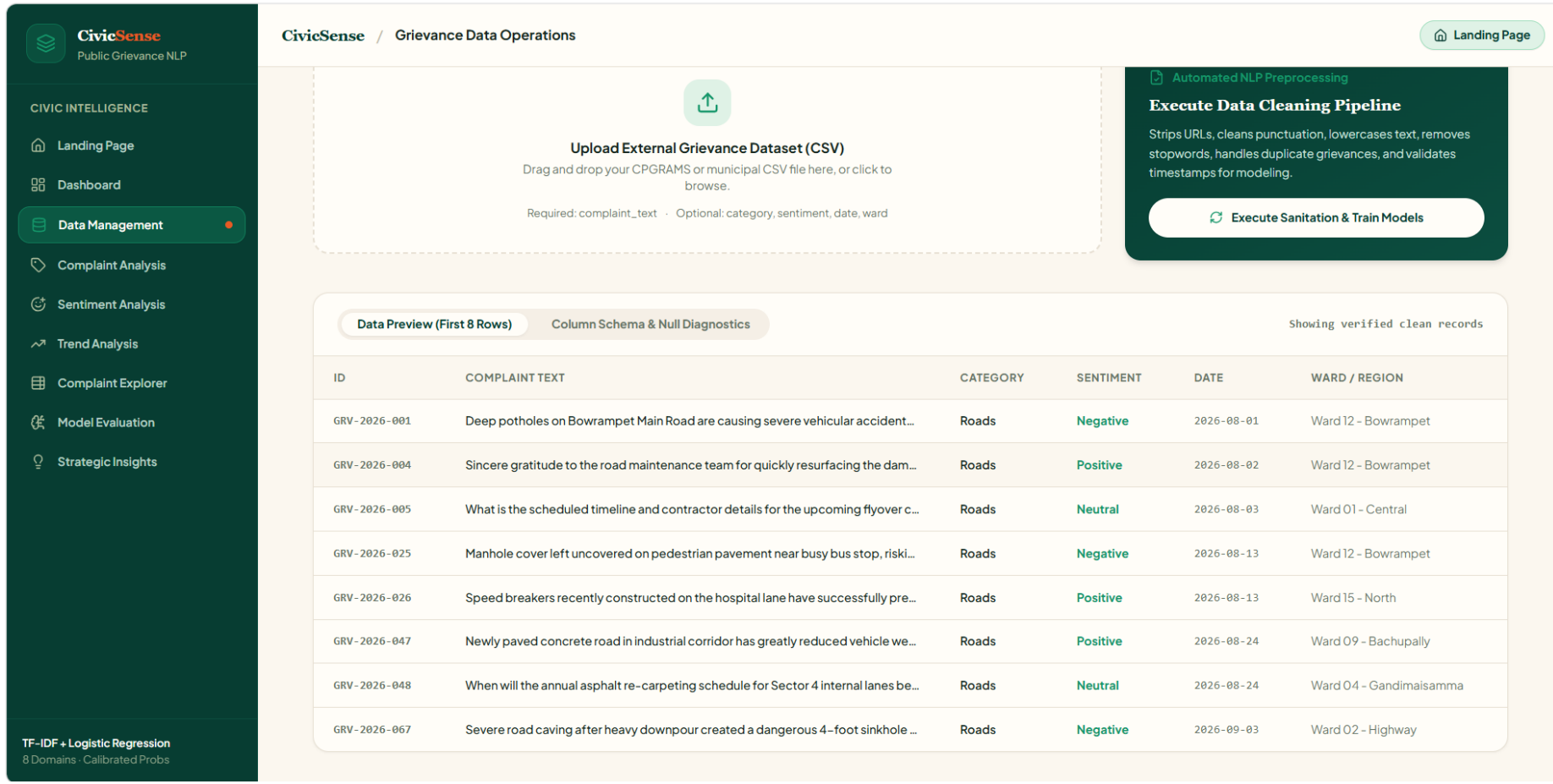
```
cleaned = complaints.apply(clean_text)
X_train, X_test, y_train, y_test = train_test_split(
   cleaned, labels, test_size=0.25,
   random_state=42, stratify=labels
)
Xtr = tfidf.fit_transform(X_train)
Xte = tfidf.transform(X_test)
model.fit(Xtr, y_train)
pred = model.predict(Xte)

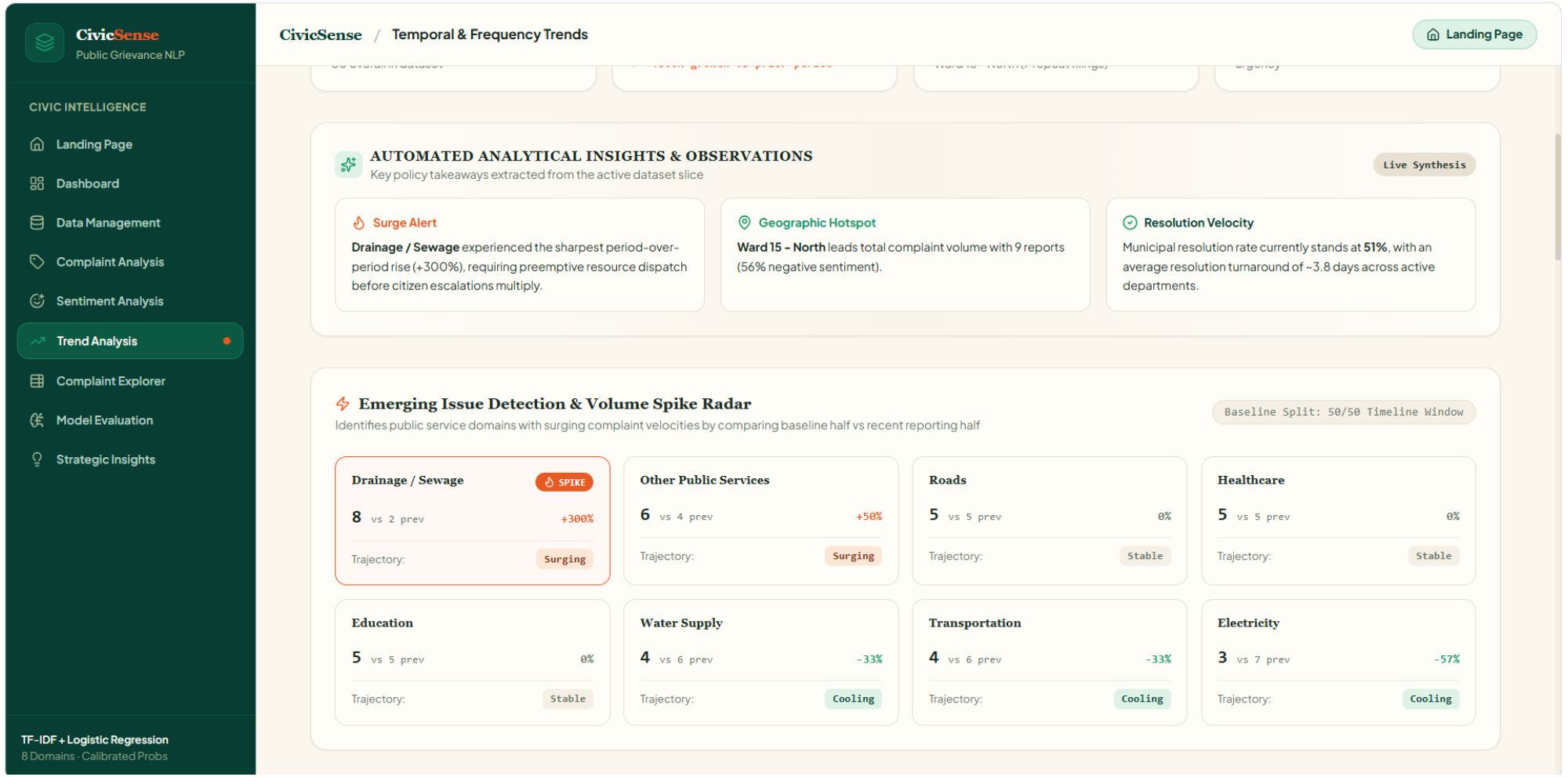
# Evaluate with accuracy, precision,
# recall, F1-score and confusion matrix.
```

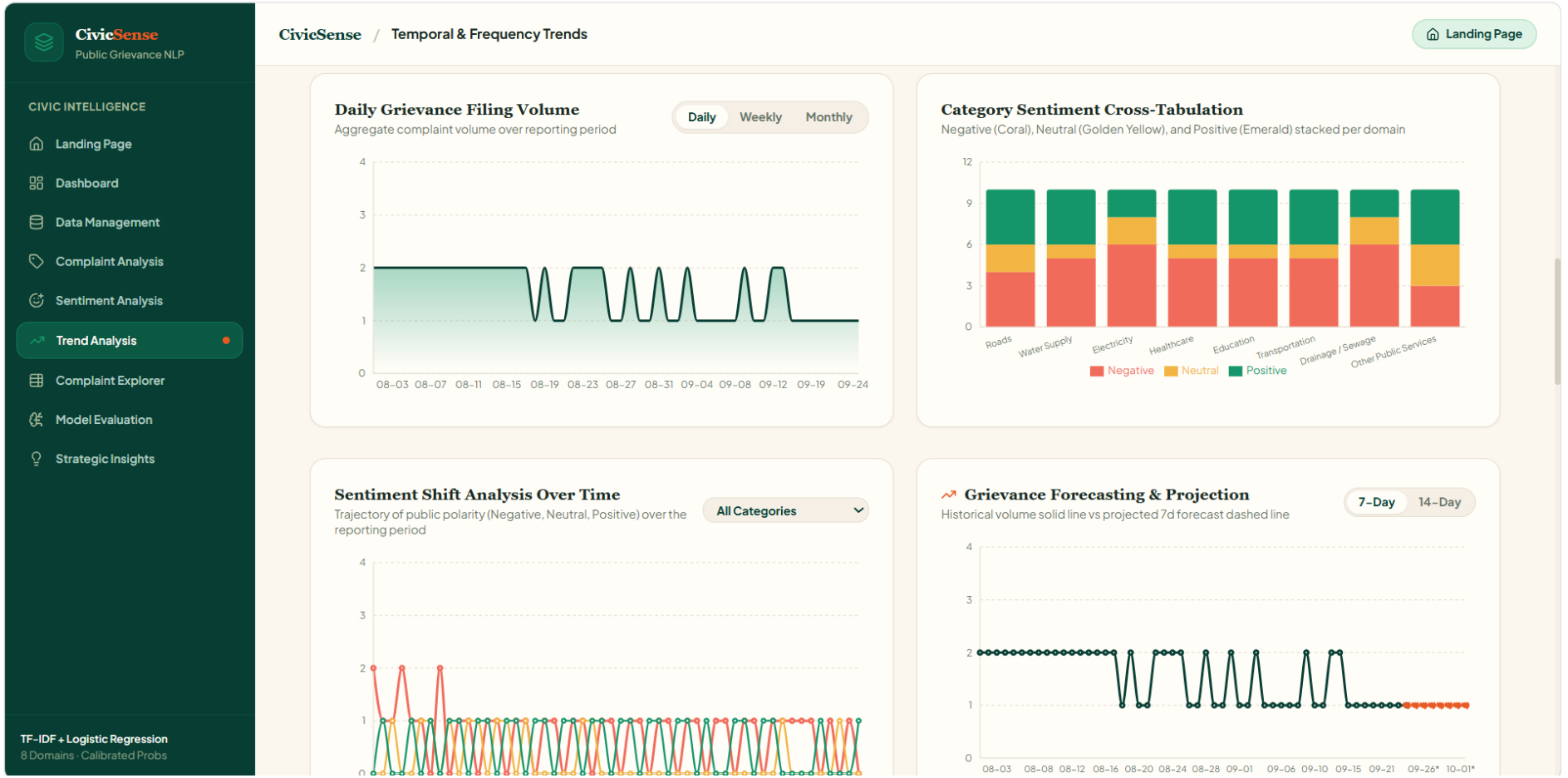
OUTPUT


localhost:3000
Chat
For quick access, place your favorites here on the favorites bar. Manage favorites now
CivicSense
Home
About
Analysis
Dashboard
Insights
Contact
Get Started
CivicSense
Government Grievance & Public Sentiment Analysis System
Roads
Water Supply
Electricity
Healthcare
Education
Transportation
FROM CITIZEN VOICES TO ACTIONABLE INSIGHTS
Explore Dashboard
Learn More
Total Grievances
80
Sentiment
Positive
Neutral
Negative
Complaint Trends
Top Complaint Domains
Roads
Water Supply
Electricity
Healthcare
Scroll for System Highlights


CivicSense
Public Grievance NLP
CIVIC INTELLIGENCE
Landing Page
Dashboard
Data Management
Complaint Analysis
Sentiment Analysis
Trend Analysis
Complaint Explorer
Model Evaluation
Strategic Insights
TF-IDF + Logistic Regression
8 Domains · Calibrated Probs
CivicSense / Executive Overview
Landing Page
Civic Intelligence & Public Sentiment System · 8 Public Service Domains
Executive Grievance & Sentiment Dashboard
Multi-domain civic diagnostics: TF-IDF feature extraction, Logistic Regression sentiment polarity, resolution SLAs, and operational priority ranking.
Classify New Grievance
Explore All (80)
Filter View:
All Categories (8)
All Sentiments
All Wards / Regions
Showing 80 of 80 records
Active Grievances
80
Click to explore records
Drainage / Sewage
10 (13%)
Sewer overflows & choke alerts
Special 8th Domain
Negative Sentiment
39 (49%)
Acute citizen dissatisfaction
Filter negative cases
Positive Feedback
28 (35%)
Repairs completed on time
Filter positive cases
Resolution Rate
51.3% (41/80)
Backlog: 39
~3.5d SLA
Sentiment Distribution
Click any slice to filter complaints
Detailed View →
Municipal Spatial Density
Interactive ward grievance pins
Live Map
Ward 3 Central
Ward 7 North
Top Complaint Domains
Click bar to filter complaints
Classifier →
Roads
10 cases
Water Supply
10 cases


CivicSense
Public Grievance NLP
CIVIC INTELLIGENCE
Landing Page
Dashboard
Data Management
Complaint Analysis
Sentiment Analysis
Trend Analysis
Complaint Explorer
Model Evaluation
Strategic Insights
TF-IDF + Logistic Regression
8 Domains · Calibrated Probs
CivicSense / Grievance Data Operations
Landing Page
Upload External Grievance Dataset (CSV)
Drag and drop your CPGRAMS or municipal CSV file here, or click to browse.
Required: complaint_text · Optional: category, sentiment, date, ward
Automated NLP Preprocessing
Execute Data Cleaning Pipeline
Strips URLs, cleans punctuation, lowercases text, removes stopwords, handles duplicate grievances, and validates timestamps for modeling.
Execute Sanitation & Train Models
Data Preview (First 8 Rows)
Column Schema & Null Diagnostics
Showing verified clean records
ID
COMPLAINT TEXT
CATEGORY
SENTIMENT
DATE
WARD / REGION
GRV-2026-001
Deep potholes on Bowrampet Main Road are causing severe vehicular accident...
Roads
Negative
2026-08-01
Ward 12 - Bowrampet
GRV-2026-004
Sincere gratitude to the road maintenance team for quickly resurfacing the dam...
Roads
Positive
2026-08-02
Ward 12 - Bowrampet
GRV-2026-005
What is the scheduled timeline and contractor details for the upcoming flyover c...
Roads
Neutral
2026-08-03
Ward 01 - Central
GRV-2026-025
Manhole cover left uncovered on pedestrian pavement near busy bus stop, riski...
Roads
Negative
2026-08-13
Ward 12 - Bowrampet
GRV-2026-026
Speed breakers recently constructed on the hospital lane have successfully pre...
Roads
Positive
2026-08-13
Ward 15 - North
GRV-2026-047
Newly paved concrete road in industrial corridor has greatly reduced vehicle we...
Roads
Positive
2026-08-24
Ward 09 - Bachupally
GRV-2026-048
When will the annual asphalt re-carpeting schedule for Sector 4 internal lanes be...
Roads
Neutral
2026-08-24
Ward 04 - Gandimaisamma
GRV-2026-067
Severe road caving after heavy downpour created a dangerous 4-foot sinkhole ...
Roads
Negative
2026-09-03
Ward 02 - Highway


CivicSense
Public Grievance NLP
CIVIC INTELLIGENCE
Landing Page
Dashboard
Data Management
Complaint Analysis
Sentiment Analysis
Trend Analysis
Complaint Explorer
Model Evaluation
Strategic Insights
TF-IDF + Logistic Regression
8 Domains · Calibrated Probs
CivicSense / Temporal & Frequency Trends
Landing Page
AUTOMATED ANALYTICAL INSIGHTS & OBSERVATIONS
Key policy takeaways extracted from the active dataset slice
Live Synthesis
Surge Alert
Drainage / Sewage experienced the sharpest period-over-period rise (+300%), requiring preemptive resource dispatch before citizen escalations multiply.
Geographic Hotspot
Ward 15 - North leads total complaint volume with 9 reports (56% negative sentiment).
Resolution Velocity
Municipal resolution rate currently stands at 51%, with an average resolution turnaround of ~3.8 days across active departments.
Emerging Issue Detection & Volume Spike Radar
Identifies public service domains with surging complaint velocities by comparing baseline half vs recent reporting half
Baseline Split: 50/50 Timeline Window
Drainage / Sewage
SPIKE
8 vs 2 prev
+300%
Trajectory:
Surging
Other Public Services
6 vs 4 prev
+50%
Trajectory:
Surging
Roads
5 vs 5 prev
0%
Trajectory:
Stable
Healthcare
5 vs 5 prev
0%
Trajectory:
Stable
Education
5 vs 5 prev
0%
Trajectory:
Stable
Water Supply
4 vs 6 prev
-33%
Trajectory:
Cooling
Transportation
4 vs 6 prev
-33%
Trajectory:
Cooling
Electricity
3 vs 7 prev
-57%
Trajectory:
Cooling


CivicSense
Public Grievance NLP
CIVIC INTELLIGENCE
Landing Page
Dashboard
Data Management
Complaint Analysis
Sentiment Analysis
Trend Analysis
Complaint Explorer
Model Evaluation
Strategic Insights
TF-IDF + Logistic Regression
8 Domains · Calibrated Probs
CivicSense / Temporal & Frequency Trends
Landing Page
Daily Grievance Filing Volume
Aggregate complaint volume over reporting period
Daily
Weekly
Monthly
4
3
2
1
0
08-03 08-07 08-11 08-15 08-19 08-23 08-27 08-31 09-04 09-08 09-12 09-19 09-24
Category Sentiment Cross-Tabulation
Negative (Coral), Neutral (Golden Yellow), and Positive (Emerald) stacked per domain
12
9
6
3
0
Roads
Water Supply
Electricity
Healthcare
Education
Transportation
Drainage / Sewage
Other Public Services
Negative
Neutral
Positive
Sentiment Shift Analysis Over Time
Trajectory of public polarity (Negative, Neutral, Positive) over the reporting period
All Categories
4
3
2
1
0
Grievance Forecasting & Projection
Historical volume solid line vs projected 7d forecast dashed line
7-Day
14-Day
4
3
2
1
0
08-03 08-08 08-12 08-16 08-20 08-24 08-28 09-01 09-06 09-10 09-15 09-21 09-26* 10-01*

Results & Performance Comparison

Metric	Category LR	Sentiment LR	NB baseline
Accuracy	[Run model]	[Run model]	[Run model]
Precision	[Run model]	[Run model]	[Run model]
Recall	[Run model]	[Run model]	[Run model]
F1-score	[Run model]	[Run model]	[Run model]

Metric	Model A	Model B	Model C
Accuracy	0.00	0.00	0.00
Precision	0.00	0.00	0.00
Recall	0.00	0.00	0.00
F1-score	0.00	0.00	0.00
ROC-AUC	0.00	0.00	0.00

Model results: populate from final run; do not report unverified scores.

Note: No verified execution metrics were included in the supplied archive. Enter actual category and sentiment scores after running evaluation.

Conclusion

1. Built CivicSense, a grievance analytics prototype using React, Python, FastAPI and Streamlit with a supplied 80-record CSV dataset.

2. Implemented eight-category complaint classification and three-class sentiment analysis using TF-IDF and Logistic Regression.

3. Delivered data quality checks, prediction modules, trend analysis, insights and API endpoints. Model scores must be added after final evaluation.



Final Outcome

80

Demo records processed

Objectives implemented: 4 / 4 (feature scope)

Future Scope & Applications

Improve Model Performance

Fine-tune vectorizers, compare models and evaluate on larger labelled datasets.

Expand Data & Languages

Add verified records and Telugu / Hindi / code-mixed text support.

Integrate Official Workflows

Add secure submission, tracking and authorized government API integration.

Scale & Secure

Add database persistence, authentication, privacy controls and cloud deployment.

Applications

- Municipal complaint trend monitoring

Department workload analysis

Citizen feedback and sentiment review

Potential users: civic researchers and authorized service administrators

References

[1] G. Salton and C. Buckley, “Term-weighting approaches in automatic text retrieval,” Information Processing & Management, 1988.

[2] T. Joachims, “Text categorization with Support Vector Machines: Learning with many relevant features,” ECML, 1998.

[3] C. D. Manning, P. Raghavan and H. Schütze, Introduction to Information Retrieval, Cambridge University Press, 2008.

[4] F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.

[5] Official documentation: Python, Pandas, Scikit-learn, FastAPI, React, Streamlit and Vite.

Project Links

GitHub: [Insert project repository URL]

Dataset: Repository demo_grievances.csv (80 rows)

Demo video: [Insert link if available]

THANK YOU

Questions & Discussion

Team: [Insert team contact]

Guide: [Guide name]

Department of Computer Science & Engineering

K L (Deemed to be) University